Math 514, F19
Exam 1

Name: Answer

Instructions:

- There are totally 6 problems.
- You must show all of your work to receive a credit for a correct answer.

Problem	Points	Score
1	15	
2	15	
3	15	
4	10	
5	20	
6	25	
Total	100	

Good Luck !

Problem 1. (15 points) Two dice are rolled. Let A be the event that the scores on the two dice are the same. Let B be the event that that the total of the scores is six or less. Find $P(A)$, $P(B)$, and $P(A|B)$.

Solution: $A = \{(1,1), (2,2), (3,3), (4,4), (5,5), (6,6)\}$

$$\Rightarrow P(A) = \frac{6}{36} = \frac{1}{6}$$

$$B = \{(1,1), (1,2), (2,1), (1,3), (2,2), (3,1), (1,4), (2,3), (3,2), (4,1), (1,5), (2,4), (3,3), (4,2), (5,1)\}$$

$$\Rightarrow P(B) = \frac{15}{36} = \frac{5}{12}$$

$$A \cap B = \{(1,1), (2,2), (3,3)\}$$

$$\Rightarrow P(A \cap B) = \frac{3}{36} = \frac{1}{12} \Rightarrow P(A|B) = \frac{P(A \cap B)}{P(B)} = \frac{\frac{1}{12}}{\frac{5}{12}} = \frac{1}{5}$$

Problem 2. (15 points) Ten people attend a meeting. Two of them do not own a car, four of them each own one car, three of them each own two cars, and one of them owns four cars. Let X be the number of cars owned by a randomly chosen person attending the meeting. Find $E[X]$, $E[X^2]$, and $\text{Var}(X)$.

Solution: $E[X] = 0 \cdot \left(\frac{2}{10}\right) + 1 \cdot \left(\frac{4}{10}\right) + 2 \cdot \left(\frac{3}{10}\right) + 4 \cdot \left(\frac{1}{10}\right)$

$$= \frac{0}{10} + \frac{4}{10} + \frac{6}{10} + \frac{4}{10} = \frac{14}{10} = \frac{7}{5}$$

$$E[X^2] = 0^2 \cdot \left(\frac{2}{10}\right) + 1^2 \cdot \left(\frac{4}{10}\right) + 2^2 \left(\frac{3}{10}\right) + 4^2 \left(\frac{1}{10}\right)$$

$$= \frac{0}{10} + \frac{4}{10} + \frac{12}{10} + \frac{16}{10} = \frac{32}{10} = \frac{16}{5}$$

$$\text{Var}(X) = E[X^2] - (E[X])^2$$

$$= \frac{16}{5} - \left(\frac{7}{5}\right)^2$$

$$= \frac{16}{5} - \frac{49}{25} = \frac{31}{25}$$

Problem 3. (15 points) In a school, 80% of students have access to the internet at home. Let X be the number of students who have the internet access at home in a group of 6 students chosen at random. Find $E(X)$, $\mathrm{Var}(X)$, and $P(X \geq 4)$.

Solution: Let $X_i = \begin{cases} 1 & \text{if the } i\text{th student has internet access} \\ 0 & \text{otherwise.} \end{cases}$

clearly, $P\{X_i = 1\} = 0.8$, $P\{X_i = 0\} = 0.2$ → a Bernulli random variable with $p = 0.8$.

$E[X_i] = 1 \times (0.8) = 0.8$, $Var(X_i) = 0.8(1-0.8) = 0.16$

set $X = \sum_{i=1}^{6} X_i$, then

$$E[X] = \sum_{i=1}^{6} E[X_i] = 6 \times (0.8) = 4.8$$

$$Var(X) = \sum_{i=1}^{6} Var(X_i) = 6 \cdot (0.16) = 0.96$$

$$P\{X \geq 5\} = P(X=4) + P(X=5) + P(X=6)$$
$$= \binom{6}{2}(0.8^4)(0.2)^2 + \binom{6}{1}(0.8)^5(0.2) + \binom{6}{0}(0.8)^6(0.2)^0$$
$$\approx 0.9011$$

Problem 4. (10 points) If $\mathrm{Cov}(X_i, X_j) = i * j$, find the covariance $\mathrm{Cov}(X_1 + X_2, X_2 + X_3)$ and the correlation $\rho(X_1 + X_2, X_2 + X_3)$.

$$Cov(X_1+X_2, X_2+X_3) = Cov(X_1, X_2) + Cov(X_1, X_3) + Cov(X_2, X_2) + Cov(X_2, X_3)$$
$$= 1\times2 + 1\times3 + 2\times2 + 2\times3$$
$$= 15$$

$$Var(X_1+X_2) = Cov(X_1+X_2, X_1+X_2) = Cov(X_1, X_1) + Cov(X_1, X_2) + Cov(X_2, X_1) + Cov(X_2, X_2)$$
$$= 1\times1 + 1\times2 + 2\times1 + 2\times2 = 9$$

$$Var(X_2+X_3) = Cov(X_2+X_3, X_2+X_3) = Cov(X_2, X_2) + Cov(X_2, X_3) + Cov(X_3, X_2) + Cov(X_3, X_3)$$
$$= 2\times2 + 2\times3 + 3\times2 + 3\times3 = 25$$

Thus $$\rho(X_1+X_2, X_2+X_3) = \frac{Cov(X_1+X_2, X_2+X_3)}{\sqrt{Var(X_1+X_2)\, Var(X_2+X_3)}} = \frac{15}{\sqrt{9} \times \sqrt{25}} = \frac{15}{3 \cdot 5} = 1$$

Problem 5. (20 points) Assume that Tom wins \$1 at probability 1/4, \$2 at 1/4 and loses \$1 at 1/2 at a certain bet. Each bet is independent. What is the approximate probability that Tom will win at least \$40 and at most \$50 in total after 200 bets?

Solution: Let X_i denote the amount of that Tom wins at the i-th bet. It holds

$$E[X_i] = 1\times\left(\frac{1}{4}\right) + 2\cdot\left(\frac{1}{4}\right) + (-1)\cdot\left(\frac{1}{2}\right) = \frac{1}{4}$$

$$\begin{aligned} Var(X_i) &= E[X_i^2] - (E[X_i])^2 \\ &= \left[1^2\times\left(\frac{1}{4}\right) + 2^2\cdot\left(\frac{1}{4}\right) + (-1)^2\cdot\left(\frac{1}{2}\right)\right] - \left(\frac{1}{4}\right)^2 \\ &= \frac{27}{16} \end{aligned}$$

Let $X = \sum_{i=1}^{200} X_i$, then $E[X] = 200\cdot\left(\frac{1}{4}\right) = 50$

$$Var(X) = 200\cdot\left(\frac{27}{16}\right) = \frac{5400}{16}$$

By CLT, $X \sim N(50, \frac{5400}{16})$

μ $\quad \sigma^2 \to \sigma = \sqrt{\frac{5400}{16}} = \frac{15}{2}\sqrt{6}$

$$\begin{aligned} P(40 \le X \le 50) &= P(X \le 50) - P(X < 40) \\ &= P\left(Z \le \frac{50-50}{\frac{15}{2}\sqrt{6}}\right) - P\left(Z < \frac{40-50}{\frac{15}{2}\sqrt{6}}\right) \\ &= P(Z \le 0) - P\left(Z < -\frac{4}{3\sqrt{6}}\right) \\ &= \Phi(0) - \left(1 - \Phi\left(\frac{4}{3\sqrt{6}}\right)\right) \\ &\approx 0.5 - (1 - \Phi(0.5443)) \\ &\approx 0.2069 \end{aligned}$$

Problem 6. (25 points) The price of a certain stock $S(t)$, $t \geq 0$ is modeled by a geometric Brownian motion process with the ***weekly*** drift parameter $\mu = 0.0006$ and volatility parameter $\sigma = 0.035$.

(a) Find the probability that the price of the stock will be up at the end of each of the next 2 weeks.

Solution: $P\left(\frac{S(1)}{S(0)} > 1\right) = P\left(\log\left(\frac{S(1)}{S(0)}\right) > 0\right) = P(X(1) > 0)$ $\quad \begin{cases} X(0) = 0, \\ X(t) \sim N(\mu t, \sigma^2 t) \end{cases}$

$P(S(1) > S(0))$ $= P\left(Z > \frac{0 - \mu \cdot 1}{\sigma \cdot \sqrt{1}}\right) = P\left(Z > \frac{0 - 0.0006}{0.035}\right)$

$\approx P(Z > -0.01711) = P(Z < 0.01711)$

$= \Phi(0.01711) \approx 0.5068$ $\quad$ equal due same time increment!

$\Rightarrow P(S(2) > S(1) > S(0)) = P(S(2) > S(1))\, P(S(1) > S(0))$

$\approx 0.5068^2 \approx 0.2569$

(b) Estimate the probability that the price of the stock will be up by at least 5% at the end of 4 weeks.

Solution: $P(S(4) > 1.05\, S(0)) = P\left(\frac{S(4)}{S(0)} > 1.05\right)$

$= P\left(\log\left(\frac{S(4)}{S(0)}\right) > \log(1.05)\right) \approx P(X(4) > 0.0488)$

$= P\left(Z > \frac{0.0488 - \mu \cdot 4}{\sigma \cdot \sqrt{4}}\right) = P\left(Z > \frac{0.0488 - (0.0006) \times 4}{(0.035) \cdot 2}\right)$

$\approx P(Z > 0.6629) = 1 - \Phi(0.6629)$

≈ 0.2537

(c) What is the expected stock price at the end of one year (≈ 52 weeks) if the initial price $S(0) = 100$.

Solution: $E[S(52)] = S(0)\, e^{\mu \times (52) + \frac{\sigma^2}{2}(52)}$

$= 100 \times e^{(0.0006) \times (52) + \frac{(0.035)^2}{2} \cdot (52)}$

$\approx 100 \times e^{0.063}$

≈ 106.51

Extra Credit. (4 points) Two cards are dealt from a deck. Find the conditional probability that one of the cards is the king of hearts given that at least one of the cards is a king.

Solution: $A = \{$One of the cards is the king of hearts$\}$.

$B = \{$At least one of the cards is a king$\}$

$A \cap B = A$.

$$P\{A\} = 1 - \frac{51}{52} \cdot \frac{50}{51} = 1 - \frac{25}{26} = \frac{1}{26}$$

$$P\{B\} = 1 - \frac{48}{52} \cdot \frac{47}{51} = 1 - \frac{188}{221} = \frac{33}{221}$$

$$\Rightarrow P\{A|B\} = \frac{P\{A \cap B\}}{P\{B\}} = \frac{\frac{1}{26}}{\frac{33}{221}} = \frac{221}{858} = \frac{17}{66}$$